



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

PRE - SCORE

28-10-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TAAS, TTAS & TEAS)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C	A,B,C,D	B,C,D	B,C,D	B	A,B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	A	B	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	5.00	6.00	2.00	4.00	2000.00	8.60	0.00	6.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	A,B,D	A,C,D	A,B,C	A,C	B		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	B	B	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.00	8.00	9.00	3.00	4.00	223.40	6.00	4.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C	C	D	B,C,D	A,B,C,D	A,B		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	A	C	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.50	1.50	0.40	0.33	0.75	0.25	0.00	2.50

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,C)

$$T = \frac{2\pi m}{Bq} = \frac{2\pi}{\alpha B_0}$$

$$\text{at } t = \frac{\pi}{\alpha B_0} = \frac{T}{2};$$

velocity of particle is $-v_0 \hat{i} + v_0 \hat{k}$ speed will always remains constant $= v_0 \sqrt{2}$

$$\text{at } t = \frac{2\pi}{\alpha B_0} = T; \text{ displacement is equal to pitch,}$$

$$\Delta x = V_0 T = \frac{2\pi V_0}{\alpha B_0}$$

$$\text{at } t = \frac{2\pi}{\alpha B_0} = T;$$

$$\text{distance} = \text{speed} \times T = \frac{2\sqrt{2}V_0\pi}{\alpha B_0}$$

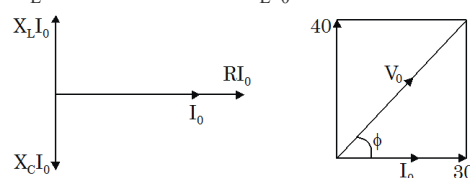
3. Ans (B,C,D)

$$\omega = \frac{2\pi}{4t_0} = \frac{5 \times 2\pi}{\pi} \times 10^3 = 10^4 \text{ rad/s}$$

$$V_C = 20 \sin \omega t \Rightarrow X_C I_0 = 20$$

$$V_R = 30 \cos \omega t \Rightarrow R I_0 = 30$$

$$V_L = -60 \sin \omega t \Rightarrow X_L I_0 = 60$$



$$V_0 = 50 \text{ Volt } \phi = \tan^{-1} \left(\frac{4}{3} \right) = 53^\circ$$

$$V = 50 \cos(\omega t + \phi) = 50 \cos \left(10000t + \frac{53\pi}{180} \right)$$

$$I_0 = \frac{50}{500} = 0.1 \text{ A} \Rightarrow X_C = 200 \Omega$$

$$\Rightarrow C = \frac{1}{X_C \omega} = \frac{1}{200 \times 10^4} = 0.5 \mu\text{F}$$

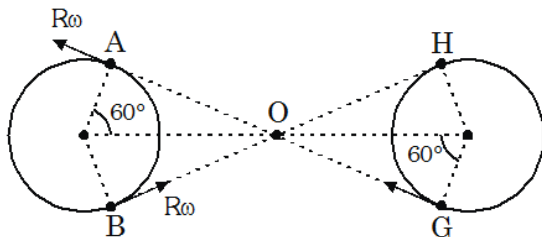
$$R = \frac{30}{0.1} = 300 \Omega, X_L = 600 \Omega$$

$$\Rightarrow L = \frac{600}{10^4} = 60 \text{ mH}$$

5. Ans (B)

If observed frequency from A is f_1 and from B is f_2 .

$$f_{1\max} = 120 \left[\frac{320}{320 - 4} \right] = 121.52$$



$$f_{1\min} = 120 \left[\frac{320}{320 + 4} \right] = 118.52 \text{ Hz}$$

$$f_{2\max} = 120 \left[\frac{320}{320 - 20} \right] = 128 \text{ Hz}$$

$$f_{2\min} = 120 \left[\frac{320}{320 + 20} \right] = 112.94 \text{ Hz}$$

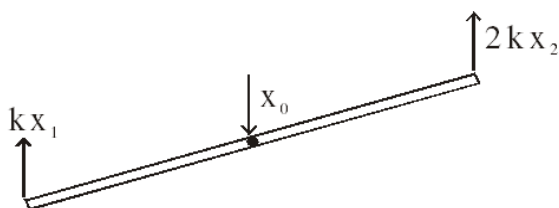
Maximum beat frequency = 9.48 Hz

Minimum beat frequency = 0

SECTION-I (ii)

8. Ans (A)

When the bead is lowered



$$x_0 = \frac{x_1 + x_2}{2} \text{ and } Kx_1 + 2Kx_2 \text{ or } x_1 = 2x_2$$

$$\therefore x_2 = \frac{2x_0}{3}$$

$$x_1 = \frac{4x_0}{3}$$

Energy of oscillation

$$= \frac{1}{2}K \left(\frac{4y_0}{3} \right)^2 + \frac{1}{2}2K \left(\frac{2y_0}{3} \right)^2$$

$$= \frac{4}{3}Ky_0^2$$

also, from mean position

$$- \left[K \frac{4y}{3} + 2K \left(\frac{2y}{3} \right) \right] = ma$$

$$a = -\frac{8K}{3m}y$$

$$T = 2\pi\sqrt{\frac{3m}{8K}}$$

9. Ans (B)

$$\sqrt{\frac{\rho v g}{\mu}} = 2\sqrt{\frac{(\rho - \rho_L)vg}{\mu}}$$

$$\rho = 4(\rho - \rho_L)$$

$$\Rightarrow \rho_L = 1.5 \text{ gm/cc}$$

10. Ans (C)

$$\frac{1}{2L} \sqrt{\frac{\rho v g - \rho_L \frac{v}{3} g}{\mu}} = f'$$

$$f = \frac{1}{2L} \sqrt{\frac{\rho v g}{\mu}}$$

$$\frac{f'}{f} = \sqrt{1 - \frac{\rho_L}{3\rho}} = \sqrt{1 - \frac{1.5}{3 \times 2}} = \frac{\sqrt{3}}{2}$$

$$f' = 280 \frac{\sqrt{3}}{2} = 140\sqrt{3} \text{ Hz}$$

SECTION-II

1. Ans (5.00)

$$P = i_{\text{rms}} V_{\text{rms}} \cos \phi$$

$$V_{\text{rms}} = 260 \text{ volt}$$

$$i_{\text{rms}} = \frac{V_{\text{rms}}}{Z} = \frac{260}{26} = 10$$

$$\text{P.F.} = \cos \phi = \frac{R}{Z} = \frac{10}{26}$$

$$\therefore Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2} = 26$$

$$\therefore P = (10)(260) \left(\frac{10}{26} \right) = 1000 \text{ W}$$

2. Ans (6.00)

$$\omega = \frac{1}{\sqrt{LC_{\text{eq}}}} = 100\pi \Rightarrow T = \frac{1}{50} \text{ sec.}$$

3. Ans (2.00)

$$\tau = MB = C\theta$$

$$\Rightarrow N i A B = C \frac{\pi}{2} \Rightarrow C = \frac{2NiAB}{\pi} \quad (1)$$

When Q is passed,

$$\int \tau dt = \int N i A B dt$$

$$I\omega = NABQ \quad (2)$$

Also, maximum deflection happens when

$$\text{entire } \frac{1}{2}I\omega^2 \text{ converts to } \frac{1}{2}C\theta_{\max}^2$$

$$\text{So, } \theta_{\max} = \sqrt{\frac{I\omega^2}{C}} = \frac{1}{\sqrt{IC}} I\omega = \frac{NABQ}{\sqrt{IC}}$$

(from (1))

$$= \frac{NABQ}{\sqrt{\frac{2NiAB}{\pi}}} = \frac{NABQ}{\sqrt{\frac{2NiAB}{\pi}}} \quad (1)$$

(from (2))

5. **Ans (2000.00)**

$$f = \frac{n}{2L} \sqrt{\frac{250}{u}} = \frac{n+1}{2L} \sqrt{\frac{160}{u}}$$

$$5N = 4n + 4$$

$$n = 4$$

$$f = \frac{4}{2 \times 2} \sqrt{\frac{250}{10^{-3}}} = 500 \text{ Hz}$$

Minimum no. of loop = 1

$$\frac{4}{2L} \sqrt{\frac{250}{u}} = \frac{1}{2L} \sqrt{\frac{Mg}{u}}$$

$$20 = \sqrt{M}$$

$$M = 400 \text{ kg}$$

6. **Ans (8.60)**

$$\beta = 10 \log \frac{I}{I_0}$$

Intensity from one source at 3r is = $\frac{I}{9}$

$$\text{From 8 coherent source} = \frac{64I}{9}$$

$$dB = 10 \log \left(\frac{64I}{9I_0} \right)$$

$$dB = \beta + 10 [\log 64 - \log 9]$$

$$\eta = 10 [6 \times 0.3 - 2 \times 0.47] = 8.60$$

7. **Ans (0.00)**

$$m \left(2 \frac{dr}{dt} \times \frac{d\theta}{dt} + r \frac{dw}{dt} \right) = q \frac{dr}{dt} B$$

$$\Rightarrow 2mr \frac{dr}{dt} \times \frac{d\theta}{dt} + r^2 \frac{dw}{dt} = qrB \frac{dr}{dt}$$

$$\Rightarrow m \frac{d}{dt} (r^2 w) = \frac{q B r dr}{dt}$$

Change in angular momentum of particle

$$= \int_{r=0}^{r=R} q B r dr = 0$$

$$\Rightarrow \frac{d\theta}{dt} = 0 \text{ at time when particle leaves the region. So velocity of particle is radially outward}$$

8. **Ans (6.00)**

$$V_L^2 + R_R^2 = 150^2 \Rightarrow V_L^2 = 150^2 - 120^2$$

$$\Rightarrow V_L = 90 \text{ V}$$

$$\text{Also } A_C^2 + V_R^2 = 130^2 \Rightarrow V_C = 50 \text{ V}$$

$$V = \sqrt{V_R^2 + (V_L - V_C)^2} = \sqrt{120^2 + 40^2} = 40\sqrt{10}$$

Power factor

$$\cos \phi = \frac{R}{Z} = \frac{V_R}{V} = \frac{120}{40\sqrt{10}} = \frac{3}{\sqrt{10}}$$

PART-2 : CHEMISTRY

SECTION-I (i)

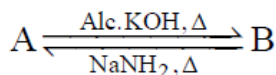
3. **Ans (A,C,D)**

[Magnesite: MgCO_3]

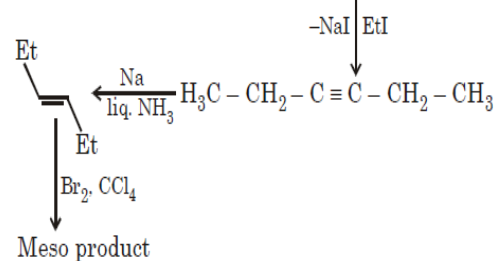
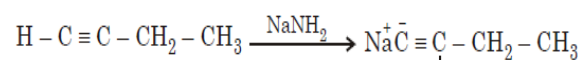
4. **Ans (A,B,C)**

In electrolytic refining impure copper rod used as anode and pure thin copper sheet used as cathode.

5. **Ans (A,C)**



6. **Ans (B)**



SECTION-I (ii)

7. **Ans (B)**

$$[\text{HCl}] = \frac{0.01}{0.1} = 0.1 \text{ M}$$

$$\text{pOH} = 4.74 + \log \left(\frac{0.3 + 0.1}{0.2 - 0.1} \right) = 5.34$$

$$\therefore \text{pH} = 14 - 5.34 = 8.66$$

SECTION-II

1. **Ans (1.00)**

$$(110 + 230 - 196.5) = (2.4 \times 10^{-6} - 1.4 \times 10^{-6})$$

$$\times \frac{10^3}{C \left(\frac{\text{mol}}{\text{litre}} \right)}$$

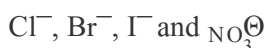
$$C = \frac{10^{-3} \text{ mol}}{143.5 \text{ litre}}$$

$$S = \frac{10^{-3}}{143.5} \times 143.5 \text{ gm} = \frac{1 \text{ mg}}{\text{litre}} = 1.00$$

4. **Ans (3.00)**



5. **Ans (4.00)**



7. **Ans (6.00)**

All type of given reactions are possible

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B,C)

$$\text{Let } G(x) = e^x \cos x f(x)$$

$$G'(x) = \cos x f'(x) + (\cos x - \sin x) f(x) \leq 0$$

$G(x)$ is a non increasing function

$$G(\pi/2) \geq G(x), \forall x \geq \pi/2$$

$$G(6) \leq 0$$

$$e^6 \cos 6 f(6) \leq 0$$

$$\Rightarrow f(6) \leq 0 \text{ but } f(x) \geq 0$$

$$\Rightarrow f(6) = 0$$

2. Ans (C)

$$\frac{dy}{dx} + \frac{1}{x} \tan y = \frac{1}{x^2} \tan y \cdot \sin y$$

$$(\cot y \operatorname{cosec} y) \frac{dy}{dx} + \frac{\cos y}{x} = \frac{1}{x^2}$$

Put $\operatorname{cosec} y = t$

$$\Rightarrow -\cot y \operatorname{cosec} y \frac{dy}{dx} = \frac{dt}{dx}$$

$$\Rightarrow \frac{-dt}{dx} + \frac{t}{x} = \frac{1}{x^2}$$

$$\Rightarrow \frac{dt}{dx} - \frac{t}{x} = \frac{-1}{x^2}$$

$$\text{I. f.} = e^{\int \frac{-1}{x} dx} = e^{-\ln x} = 1/x$$

$$\Rightarrow \frac{t}{x} = \int \frac{-1}{x^3} dx = \frac{1}{2x^2} + c$$

$$\Rightarrow t = \frac{1}{2x} + cx$$

$$\frac{1}{\sin y} = \frac{1 + 2cx^2}{2x}$$

$$\Rightarrow 2x = \sin y (1 + 2cx^2)$$

3. Ans (D)

$$\int \left(\frac{2e^{\tan^{-1}x+x^2}}{I} \cdot \frac{(x)}{II} + x^2 \left(\frac{2x^3+2x+1}{x^2+1} \right) e^{\tan^{-1}x+x^2} \right) dx$$

$$2 \left[e^{\tan^{-1}x+x^2} \left(\frac{x^2}{2} \right) - \int e^{\tan^{-1}x+x^2} \left(\frac{2x^3+2x+1}{x^2+1} \right) \frac{x^2}{2} dx \right]$$

$$+ \int x^2 \frac{(2x^3+2x+1)}{x^2+1} e^{\tan^{-1}x+x^2} dx$$

$$= x^2 e^{\tan^{-1}x+x^2} + C$$

4. Ans (B,C,D)

$$f(x) = \begin{cases} \lim_{n \rightarrow \infty} \frac{\left(1 + \left[\frac{\sin x}{x}\right]^{2n}\right) \{x\}}{1 + \left[\frac{x}{\tan x}\right]^{2n}} & ; 0 < x < \frac{\pi}{2} \\ \frac{\{x\}^3 - 3\{x\}}{4 - [x]} & ; -2 < x \leq 0 \\ x^3 - 27x - 46 & ; x \leq -2 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} \{x\} & ; 0 < x < \frac{\pi}{2} \\ \frac{\{x\}^3 - 3\{x\}}{4 - [x]} & ; -2 < x \leq 0 \\ x^3 - 27x - 46 & ; x \leq -2 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} x^3 - 27x - 46 & ; x \leq -2 \\ \frac{(x+2)^3 - 3(x+2)}{6} & ; -2 < x < -1 \\ \frac{(x+1)^3 - 3(x+1)}{5} & ; -1 \leq x < 0 \\ 0 & ; x = 0 \\ x & ; 0 < x < 1 \\ x - 1 & ; 1 \leq x < \frac{\pi}{2} \end{cases}$$

Now conclude the options

5. Ans (A,B,C,D)

$f'(x)$ must be of least degree so take

$$f'(x) = a(x-1)(x-3)$$

$$\Rightarrow f(x) = \left(\frac{a}{3}\right) x^3 - 2ax^2 + 3ax + c. \text{ Using}$$

$$f(1) = 6 \text{ and } f(3) = 2 \text{ we get } a = 3, c = 2$$

$$\therefore f(x) = x^3 - 6x^2 + 9x + 2$$

6. Ans (A,B)

$$f'(x) - 3x^2 f(x) > 0 \forall x \geq 1$$

$$\Rightarrow \frac{d}{dx} (e^{-x^3} f(x)) > 0 \forall x \geq 1$$

$$\Rightarrow e^{-x^3} f(x) \geq e^{-1} f(1) \Rightarrow e^{-x^3} f(x) \geq e$$

$$\Rightarrow f(x) \geq e^{x^3+1} \forall x \geq 1$$

$$f(x) \geq e^2$$

SECTION-I (ii)

7. Ans (D)

we have, $f(x) = (x-a)^3 + b \dots(i)$

Since, Rolle's theorem is applicable to $g(x)$ at $x = 2$.

So, $g(a) = g(b)$ and $g'(2) = 0$

$$\Rightarrow 0 = f(b) - f(a) + (a-b) \cdot f'(b) + 3(b-a)^2$$

$$\text{and } (a-2)f''(2) + 6(2-a) = 0$$

$$\therefore f'(b) = \frac{f(b) - f(a)}{(b-a)} + 3(b-a) \dots(ii)$$

$$\text{and } f''(2) = 6, (a \neq 2) \dots(iii)$$

$$\text{from (i), } f''(x) = 6(x-a) \text{ and } f''(2) = 6$$

$$\Rightarrow 6(2-a) = 6 \Rightarrow a = 1$$

also, from (i) and (ii)

$$(b-a) = \frac{3}{2} \Rightarrow b = \frac{5}{2}$$

$$\therefore \int_1^{5/2} \left((x-1)^3 + \frac{5}{2} \right) dx = \frac{321}{64}$$

8. Ans (A)

we have, $f(x) = (x-a)^3 + b \dots(i)$

Since, Rolle's theorem is applicable to $g(x)$ at $x = 2$.

So, $g(a) = g(b)$ and $g'(2) = 0$

$$\Rightarrow 0 = f(b) - f(a) + (a-b) \cdot f'(b) + 3(b-a)^2$$

$$\text{and } (a-2)f''(2) + 6(2-a) = 0$$

$$\therefore f'(b) = \frac{f(b) - f(a)}{(b-a)} + 3(b-a) \dots(ii)$$

$$\text{and } f''(2) = 6, (a \neq 2) \dots(iii)$$

$$\text{from (i), } f''(x) = 6(x-a) \text{ and } f''(2) = 6$$

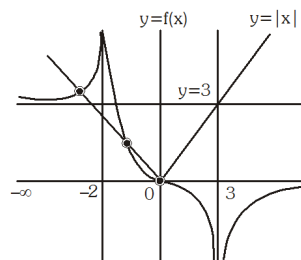
$$\Rightarrow 6(2-a) = 6 \Rightarrow a = 1$$

also, from (i) and (ii)

$$(b-a) = \frac{3}{2} \Rightarrow b = \frac{5}{2}$$

$$\therefore \int_1^{5/2} \left((x-1)^3 + \frac{5}{2} \right) dx = \frac{321}{64}$$

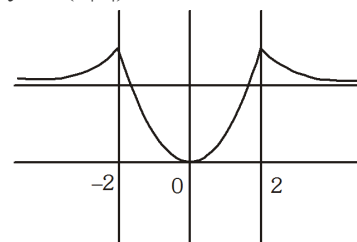
9. Ans (C)



Maximum possible number of solutions = 3

10. Ans (A)

$$y = f(-|x|)$$



Clearly non-differentiable at 2 points.

SECTION-II

1. Ans (0.50)

$$\text{put } x = \frac{1}{t},$$

$$\text{So } I = \int_0^1 \ln \left(\frac{1}{t(1-t)} \right) dt$$

$$= - \int_0^1 \ln t \, dt - \int_0^1 \ln(1-t) \, dt$$

$$= -2 \int_0^1 \ln t \, dt = 2$$

2. Ans (1.50)

$$\text{Since, } g: \mathbb{R} \rightarrow \{4\}$$

$$\Rightarrow g(x) \text{ is constant function equal to 4.}$$

$$\Rightarrow g(x) = 4$$

$$\Rightarrow x^3(f'(t) - 2) + x^2f''(t) + 4x(f(0) + 6) + 4 = 4, \forall x \in \mathbb{R}$$

$$\therefore f'(t) - 2 = 0, f''(t) = 0, f(0) + 6 = 0$$

$$f(t) = 2t - 6$$

$$\Rightarrow h(x) = \begin{cases} -x^2 + 8x & ; 0 \leq x < 4 \\ x^2 - 8x + 32 & ; 4 \leq x \leq 6 \\ (x-6)^2 + 20 & ; 6 \leq x \leq 12 \end{cases}$$

$$\therefore \text{Range of } h(x) \text{ is } [0, 56]$$

$$\text{i.e. } N = 57$$

3. Ans (0.40)

Let $y = mx + c$ is a bitangent, therefore the equation $x^4 + 2x^3 - 11x^2 - 13x + 35 = mx + c$ must have exactly two real solutions.

therefore the expression

$x^4 + 2x^3 - 11x^2 - (13+m)x + (35-c)$ is the product of the squares of two linear factors, so perfect square of a quadratic expression.

$$\begin{aligned} & \text{Let } x^4 + 2x^3 - 11x^2 - (13+m)x + (35-c) \\ &= (x^2 + px + q)^2 \end{aligned}$$

comparing coefficient of x^3 , $2p = 2 \Rightarrow p = 1$

comparing coefficient of x^2 , $p^2 + 2q = -11$

$$\Rightarrow q = -6$$

$$\therefore (x^2 + px + q)^2 = (x^2 + x - 6)^2$$

$$\therefore m = c = 1 \quad \therefore y = -x - 1 \text{ is bitangent}$$

$$\therefore |m + c| = 2.$$

4. Ans (0.33)

Point of intersection $\equiv (1,0)$

$$y = x^x \log_e x$$

$$y' = \frac{x^x}{x} + x^x (1 + \log_e x) \cdot \log_e x \Rightarrow m_1 = 1$$

$$y = \frac{2^x - 2}{\log_e 2} \Rightarrow y' = \frac{2^x \log_e 2}{\log_e 2} \Rightarrow m_2 = 2$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan \theta = \frac{1}{3}$$

5. Ans (0.75)

$$I_n = \int_0^{\frac{3\pi}{2}} (\ln |\sin x|) \cos(2nx) dx$$

applying integration by parts

$$I_n = \left\{ \ln |\sin x| \cdot \frac{\sin 2nx}{2n} \right\}_0^{\frac{3\pi}{2}} - \int_0^{\frac{3\pi}{2}} \frac{\cot x \cdot \sin 2nx}{2n} dx$$

$$I_n = 0 - \frac{1}{2n} I_n'$$

$$I_n' = \int_0^{\frac{3\pi}{2}} \frac{\cos x \cdot \sin 2nx}{\sin x} dx$$

$$I_n' - I_{n-1}' = \int_0^{\frac{3\pi}{2}} \frac{\cos x (\sin 2nx - \sin(2n-2)x)}{\sin x} dx$$

$$= \int_0^{\frac{3\pi}{2}} \frac{2 \cos x \cdot \cos(2n-1)x \sin x}{\sin x} dx$$

$$I_n' - I_{n-1}' = \int_0^{\frac{3\pi}{2}} 2 \cos(2n-1)x \cdot \cos x dx = 0$$

$$I_n' = I_{n-1}' = I_{n-2}' = \dots = I_1'$$

$$I_n' = \int_0^{\frac{3\pi}{2}} \frac{\sin 2x \cos x}{\sin x} dx$$

$$= \int_0^{\frac{3\pi}{2}} 2 \cos^2 x dx = \int_0^{\frac{3\pi}{2}} (1 + \cos 2x) dx$$

$$= \frac{3\pi}{2}$$

$$I_n = -\frac{3\pi}{4n}$$

$$12I_3 = -3\pi$$

$$16I_2 = -6\pi$$

$$\Rightarrow 12I_3 - 16I_2 = 3\pi$$

6. **Ans (0.25)**

$$\text{Let } f(x) = x^{\frac{1}{x}}$$

$$\Rightarrow f'(x) = x^{\frac{1}{x}} \left(\frac{\ln x - 1}{x^2} \right)$$

If $x > e$ then $f(x)$ is decreasing

$$\because 2018 > 2017 \Rightarrow f(2018) < f(2017)$$

$$\Rightarrow (2018)^{\frac{1}{2018}} < (2017)^{\frac{1}{2017}}$$

$$\Rightarrow (2018)^{2017} < (2017)^{2018}$$

$$\begin{aligned} \because \lim_{n \rightarrow \infty} \left\{ \left((2017)^{2018} \right)^n + \left((2018)^{2017} \right)^n \right\}^{\frac{1}{n}} \\ = \lim_{n \rightarrow \infty} (2017)^{2018} \left\{ 1 + \left(\frac{(2018)^{2017}}{(2017)^{2018}} \right)^n \right\}^{\frac{1}{n}} = (2017)^{2018} \end{aligned}$$

7. **Ans (0.00)**

Since, n is even, let $n = 2\ell$

Now,

$$1 + 2x + 3x^2 + 4x^3 + 5x^4 + 6x^5 + 7x^6 + 8x^7 + \dots + (2\ell + 1)x^{2\ell}$$

$$= (1 + 2x + x^2) + 2x^2(1 + 2x + x^2) + 3x^4(1 + 2x + x^2)$$

$$+ 4x^6(1 + 2x + x^2) + \dots + \ell x^{2\ell-2}(1 + 2x + x^2) + (\ell + 1)x^{2\ell}$$

$$= (1 + x)^2 (1 + 2x^2 + 3x^4 + 4x^6 + \dots + \ell x^{2\ell-2}) + (\ell + 1)x^{2\ell}$$

Which can never be zero for any value of x .

8. **Ans (2.50)**

$$\frac{y''}{y'} = \frac{2y'}{y} \Rightarrow \frac{y'}{y^2} = k$$

$$\int \frac{dy}{y^2} = \int k dx \Rightarrow -\frac{1}{y} = kx + \lambda \quad \dots(1)$$

since $(2,2)$ and $\left(8, \frac{1}{2}\right)$ lies on (1), we get C_1 :

$$xy = 4$$

shortest distance between

$$C_1 : xy = 4 \text{ and } C_2 : x^2 + y^2 = 4 \text{ is}$$

$$(2\sqrt{2} - 2) = (\sqrt{8} - 2)$$

Hence, $(a + b) = 10$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

PRE - SCORE

28-10-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TAAS, TTAS & TEAS)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	B,C,D	A,C	B,C	B	B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	B	A	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.50	6.00	0.25	4.00	1.00	4.20	3.00	8.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	A,D	B,C	A,C,D	A,B,C,D	A		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	C	C	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.00	8.00	4.00	6.00	5.00	5.00	5.00	5.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	C,D	A,B,D	A,C,D	A,B,C,D	A,B,C	A,B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	D	B	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.33	0.50	3.00	4.00	1.50	4.90	0.80	7.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B,C,D)

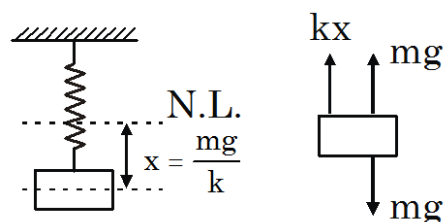
$$r < R$$

$$E \cdot 2\pi r = \frac{d}{dt} [\mu_0 n C t \times \pi r^2]$$

$$r > R$$

$$E \cdot 2\pi R = \frac{d}{dt} [\mu_0 n C t \times \pi R^2]$$

2. Ans (B,C,D)



Now as lift starts descending by acceleration 'g' of downward, in the frame of lift

$$\text{Net force} = -kx$$

$$\frac{F}{m} = \frac{-k}{m} \cdot x$$

$$t = 2\pi \sqrt{\frac{m}{k}};$$

Minimum potential energy is at the mean position = 0 when $x = 0$.

3. Ans (A,C)

(A) In standing waves phase difference between 2 points can be only zero or π .

(B) The energy of small element of string at 'O' keeps on changing with time.

(C) It is the property of standing waves.

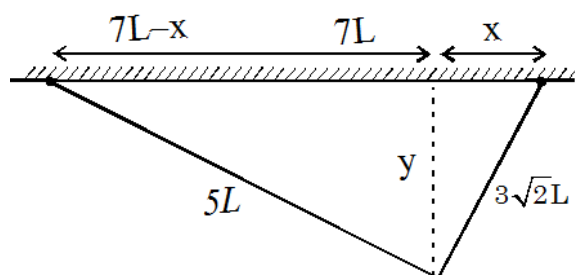
(D) $y(x, t) = A \sin kx \cos \omega t$.

$$\frac{dy(x, t)}{dt} = -A\omega \sin kx \sin \omega t$$

There will be only 2 points between A & O which will have same speed.

4. Ans (B,C)

Period of oscillation will depend on M.



$$25L^2 = (7L - x)^2 + y^2$$

$$18L^2 = y^2 + x^2$$

$$7L^2 = 4xL^2 - 14xL$$

$$14x = 42L \quad x = 3L$$

$$y = \sqrt{18L^2 - 9L^2} = 3L$$

$$2\pi\sqrt{\frac{3\ell}{g}}$$

5. Ans (B)

$$V_R = 60 = I(120)$$

$$I = \frac{1}{2}$$

$$V_L = IX_L = \frac{1}{2}X_C = 40$$

$$\omega L = 80$$

$$L = \frac{80}{4 \times 10^5} \Rightarrow 20 \times 10^{-5} = 0.2\text{mH}$$

As the circuit is in resonance

$$X_C - X_L$$

$$\frac{1}{\omega C} = \omega C$$

$$C = \frac{1}{\omega^2 L} \Rightarrow \frac{1}{4 \times 10^5 \times 80}$$

$$C = \frac{10^{-5}}{320} = \frac{1}{32} \mu\text{F}$$

$$\tan \phi = \frac{X_L - X_C}{R}$$

$$\text{If } \omega = 8 \times 10^5$$

$$X_L - X_C = 8 \times 10^5 \times 0.2 \times 10^{-3} - \frac{1(32)}{8 \times 10^5 \times 10^{-6}}$$

$$\Rightarrow 160 - 40 = 120$$

6. Ans (B,C)

$$\text{At } t = \frac{4}{3}\text{s}, Y_1 = \frac{5}{(3x - \frac{16}{3})^2 + 2} \text{ and}$$

$$Y_2 = \frac{-5}{(3x - \frac{2}{3})^2 + 2}$$

$$Y_1 + Y_2 \neq 0$$

$$\text{At } t = \frac{3}{4}\text{s}, Y_1 = \frac{5}{(3x - 3)^2 + 2} \text{ and}$$

$$Y_2 = \frac{-5}{(3x - 3)^2 + 2}$$

$$Y_1 + Y_2 = 0$$

$$\text{At } x = 1, Y_1 = \frac{5}{(3 - 4t)^2 + 2} \text{ and}$$

$$Y_2 = \frac{-5}{(-3 + 4t)^2 + 2}$$

$$Y_1 + Y_2 = 0$$

SECTION-I (ii)

7. Ans (B)

$$\text{K.E.} = \frac{1}{2}m\omega^2(A^2 - x^2)$$

$$\text{P.E.} = \frac{1}{2}m\omega^2x^2$$

8. **Ans (B)**

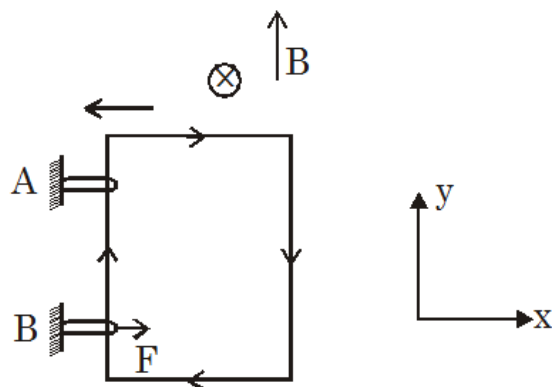
(A) and (B) B is displacement node
 $\Rightarrow a = 0, v = 0$ and $E = 0$ and deformation maximum $\Rightarrow PE$ max.

(D) B is displacement antinodes
 $\Rightarrow a = \max, KE = 0, PE = 0, V = 0$

9. **Ans (A)**

$$\vec{F}_m = q(\vec{v} \times \vec{B}) \text{ and } \vec{F}_e = q\vec{E}$$

10. **Ans (A)**



$$\tau_{\text{net}} = 0 = 10 \times \frac{1}{2} - 2 \times 4 - F \times 1$$

about A $F = -3$

SECTION-II

1. **Ans (0.50)**

$$A = 5 \text{ cm}$$

$$kx_0 = mg$$

$$k = \frac{1.8}{1/4} = 7.2 \text{ N/m.}$$

$$T = 2\pi\sqrt{\frac{0.18}{7.20}} = 1 \text{ sec.}$$

$$t = \frac{T}{2} = 0.5 \text{ sec.}$$

2. **Ans (6.00)**

At half power frequencies,

$$Z = R\sqrt{2} \Rightarrow |X_L - X_C| = R$$

$$\left(\omega L - \frac{1}{\omega C}\right)^2 = R^2 \Rightarrow \omega^2 L^2 + \frac{1}{\omega^2 C^2} - \frac{2L}{C} = R^2$$

$$\omega^4 L^2 C^2 - \omega^2 \left(\frac{2L}{C} + R^2\right) C^2 + 1 = 0$$

The product of roots is

$$\omega_1^2 \omega_2^2 = \frac{1}{L^2 C^2} = \omega_1 \omega_2 = \frac{1}{LC} = \omega_0^2$$

$$\Rightarrow \omega_0 = \sqrt{\omega_1 \omega_2} = 60 \text{ rad/s} \Rightarrow \omega_0/10 = 6$$

3. **Ans (0.25)**

$$i = \frac{1}{L} \int v dt = \frac{1}{2} \times 0 + \frac{1}{0.2} \times 0.05 \times 1 = 0.25 \text{ A.}$$

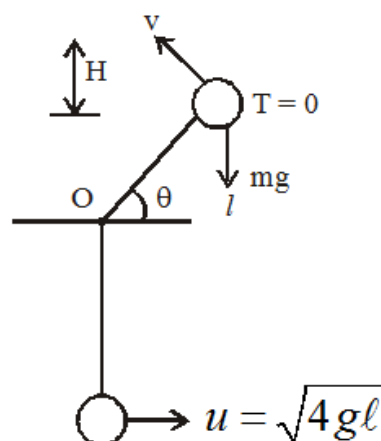
4. **Ans (4.00)**

$$T = 2\pi\sqrt{\frac{I}{C}} = 2\pi\sqrt{\frac{\frac{2}{5}mR^2}{C}} = 4 \text{ sec}$$

5. **Ans (1.00)**

$$mg \cos(90 - \theta) = \frac{mv^2}{\ell} \quad (1)$$

$$\frac{1}{2}mu^2 - \frac{1}{2}mv^2 = mg\ell(1 + \cos\theta) \quad (2)$$



$$u = \sqrt{4g\ell}$$

solving, $\sin\theta = \frac{2}{3}$

$$\text{for projectile motion, } H = \frac{v^2 \sin^2(90 - \theta)}{2g}$$

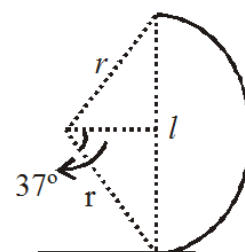
$$H = \frac{5\ell}{27}$$

$$H_{\text{max}} \text{ for ground} = \ell + \ell \sin\theta + \frac{5\ell}{27}$$

$$= \frac{50\ell}{27}$$

$$\text{for } \ell = 54 \text{ cm, } H_{\text{max}} = 1 \text{ m}$$

6. **Ans (4.20)**



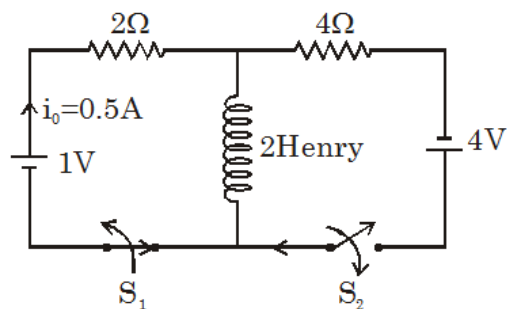
$$e = B\ell V$$

$$= B \cdot (2r \sin 37^\circ) \cdot V$$

$$= 5 \times 2 \times 1 \times \frac{3}{5} \times 0.7$$

$$= 4.20$$

7. Ans (3.00)



$$i_0 = 0.5 \text{ A}$$

When; S_2 is closed;

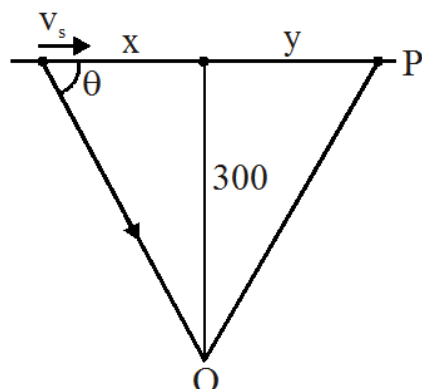
$$-L \frac{di}{dt} - 4i + 4 = 0$$

$$-2 \frac{di}{dt} - 4i_0 + 4 = 0$$

$$-2 \frac{di}{dt} + 4 \times 0.5 + 4 = 0 \quad [\because i_0 = -0.5 \text{ A}]$$

$$\frac{di}{dt} = 3$$

8. Ans (8.00)



$$3f_0 = f_0 \frac{v}{v - v_s \cos \theta}$$

$$\Rightarrow \frac{v_s}{v} \cos \theta = \frac{2}{3} \quad \dots (1)$$

$$\frac{x}{v_s} = \frac{x \sec \theta}{v} \Rightarrow \frac{v_s}{v} = \cos \theta \quad \dots (2)$$

(1) and (2)

$$\Rightarrow \cos \theta = \sqrt{\frac{2}{3}} = \frac{v_s}{v}$$

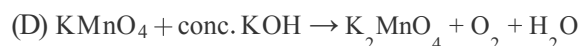
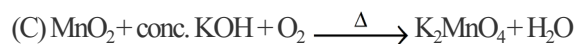
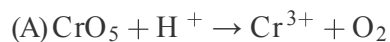
$$\frac{y}{v} = \frac{300}{v_s} \Rightarrow y = 300 \sqrt{\frac{2}{3}} = 100 \sqrt{6} \text{ m}$$

$$OP = \sqrt{(100\sqrt{6})^2 + (300)^2} = 100\sqrt{15} \text{ m}$$

PART-2 : CHEMISTRY

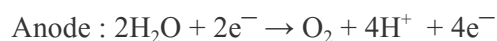
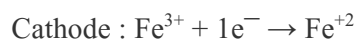
SECTION-I (ii)

8. Ans (C)



SECTION-II

1. Ans (1.00)



milliequivalent of $\text{Fe}^{2+} = \text{KMnO}_4$ in acidic medium

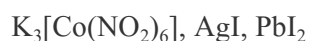
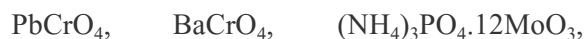
$$= 0.01 \times 30 \times 5$$

millimoles of Fe^{2+} formed = 1.5

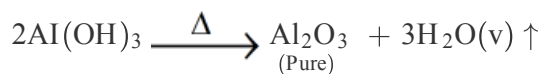
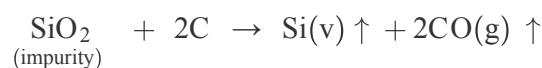
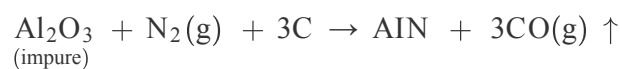
$$\text{moles of } \text{Fe}^{+2} \times 1 = \frac{I \times t}{96000} = \frac{40 \times 10^{-3} \times t}{96000}$$

$$t = 3600 \text{ sec.} = 1 \text{ hr.}$$

4. Ans (6.00)



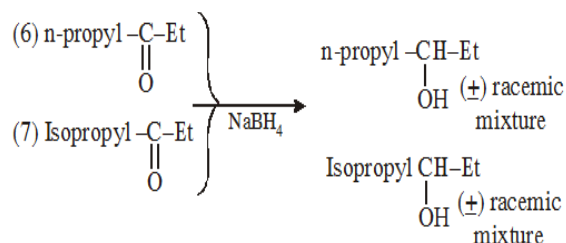
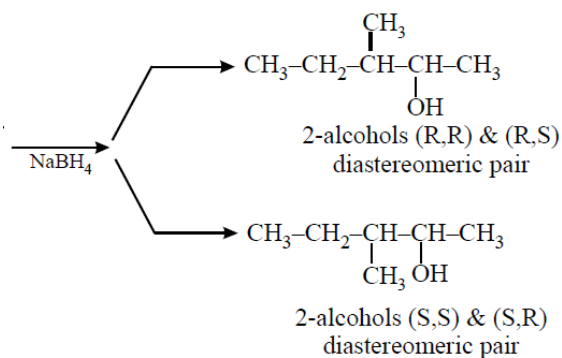
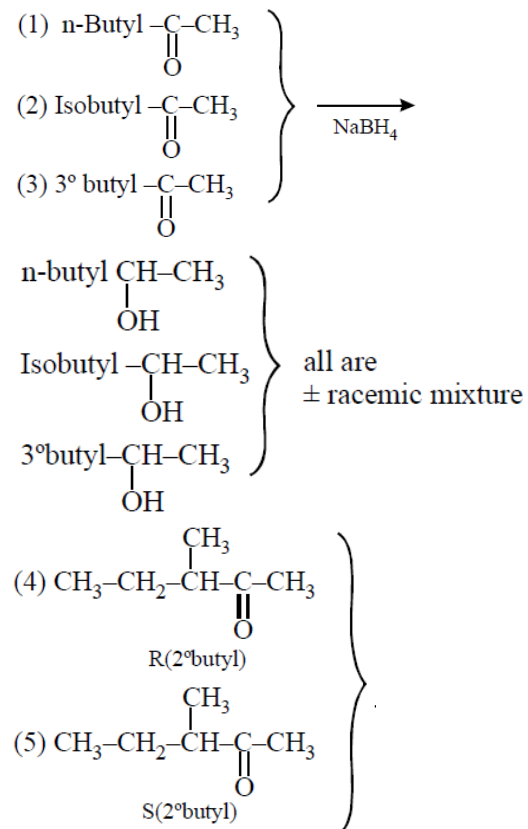
5. Ans (5.00)



7. Ans (5.00)

M. wt 100 of ketone

So m. formula = $C_6H_{12}O$



(1 ; 2 ; 3 ; 6 ; 7)

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (C,D)

Take two Cases

Case-I : When a is positive

$$a + \frac{1}{a} = x$$

$$\Rightarrow \left(a^3 + \frac{1}{a^3}\right) - 6 \left(a^2 + \frac{1}{a^2}\right) + 12 \left(a + \frac{1}{a}\right) + b \geq 0$$

$$\Rightarrow (x^3 - 3x) - 6(x^2 - 2) + 12(x) + b \geq 0; x \geq 2$$

$$\Rightarrow x^3 - 6x^2 + 9x + (12 + b) \geq 0$$

$$\text{let, } f(x) = x^3 - 6x^2 + 9x + 12$$

$$f'(x) = 0 \Rightarrow x = 1, 3$$

$$b \geq -12$$

Case-II : When a is negative

$$a + \frac{1}{a} = x$$

$$\Rightarrow \left(a^3 + \frac{1}{a^3}\right) - 6 \left(a^2 + \frac{1}{a^2}\right) + 12 \left(a + \frac{1}{a}\right) + b \leq 0$$

$$\Rightarrow (x^3 - 3x) - 6(x^2 - 2) + 12(x) + b \leq 0; x \leq -2$$

$$\Rightarrow x^3 - 6x^2 + 9x + (12 + b) \leq 0$$

$$\text{let, } f(x) = x^3 - 6x^2 + 9x + 12$$

$$b - 38 \leq 0$$

$$\therefore b \in [-12, 38], \forall |x| \geq 2$$

$$\Rightarrow q - p + 12 = 38 + 12 + 12 = 62$$

2. Ans (A,B,D)

$$I(a, b) = \int_0^{\infty} \frac{e^{-ax} - e^{-bx}}{x} dx$$

$$I(2a, 2b) = \int_0^{\infty} \frac{e^{-2ax} - e^{-2bx}}{x} dx$$

In $I(a, b)$ Put $x = 2t$ $\frac{dx}{dt} = 2$

$$I(a, b) = \int_0^{\infty} \frac{e^{-2at} - e^{-2bt}}{2t} 2dt = I(2a, 2b)$$

$$I(a, b) = \int_0^{\infty} \frac{e^{-ax}}{x} dx - \int_0^{\infty} \frac{e^{-bx}}{x} dx$$

Put $x = at$ Put $x = bt$

$$I(a, b) = \int_0^{\infty} \frac{e^{-a^2t}}{at} adt - \int_0^{\infty} \frac{e^{-b^2t}}{bt} bdt$$

$$I(a, b) = \int_0^{\infty} \frac{e^{-a^2t}}{t} dt - \int_0^{\infty} \frac{e^{-b^2t}}{t} dt$$

$$I(a, b) = \int_0^{\infty} \frac{e^{-a^2t} - e^{-b^2t}}{t} dt = I(a^2, b^2)$$

3. Ans (A,C,D)

$$= \int_0^1 (f(x))^2 dx + \int_0^1 (x^2 - x) h'(x) dx$$

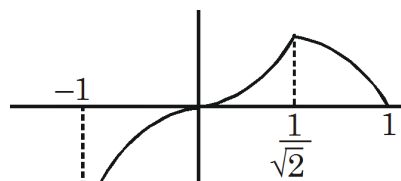
$$= \int_0^1 (f(x))^2 dx + (x^2 - x) h(x) \Big|_0^1 - \int_0^1 (2x - 1) (h(x)) dx$$

$$= \int_0^1 (f(x))^2 dx - \int_0^1 (2x - 1) (f(x) - 9x + 4) dx$$

after solving the above expression, we get

$$= 1 + \int_0^1 (f(x))^2 dx \quad \left(\because \int_0^1 (f(x))^2 dx > 0 \right)$$

4. Ans (A,B,C,D)



$$h(x) = \begin{cases} \sin^{-1} x & , -1 \leq x \leq \frac{1}{\sqrt{2}} \\ \cos^{-1} x & , \frac{1}{\sqrt{2}} \leq x \leq 1 \end{cases}$$

5. Ans (A,B,C)

$$f(x) = \begin{cases} \frac{ax^3 + bx^2 + cx + d}{x} & \forall x \neq 0 \\ 2 & x = 0 \end{cases}$$

Since $f(x)$ is continuous function therefore

$$\lim_{x \rightarrow 0} f(x) = 2$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{ax^3 + bx^2 + cx + d}{x} = 2$$

$$\Rightarrow d = 0 \text{ \& } c = 2$$

$$\Rightarrow a = 6, b = 4, c = 2, d = 0$$

$$\Rightarrow f(x) = 6x^2 + 4x + 2$$

$$\text{therefore } p = \frac{-1}{3} \text{ and } f(p) = \frac{4}{3}$$

6. Ans (A,B,D)

$$\text{Let } f(x) = (x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_{2018})$$

$$f(0) = f(1)$$

$$\Rightarrow \alpha_1 \cdot \alpha_2 \dots \alpha_{2018} = (1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_{2018})$$

$$\Rightarrow \alpha_1^2 \cdot \alpha_2^2 \dots \alpha_{2018}^2 = \alpha_1(1 - \alpha_1) \cdot \alpha_2(1 - \alpha_2) \dots$$

$$\alpha_{2018}(1 - \alpha_{2018})$$

Apply A.M. $\geq$ G.M.

$$\text{in } \alpha_1 \text{ and } 1 - \alpha_1$$

$$\frac{\alpha_1 + 1 - \alpha_1}{2} \geq (\alpha_1(1 - \alpha_1))^{\frac{1}{2}}$$

$$\Rightarrow (\alpha_1(1 - \alpha_1))^{\frac{1}{2}} \leq \frac{1}{2}$$

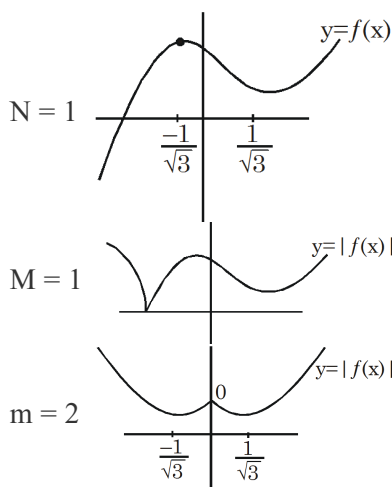
$$\Rightarrow$$

$$(\alpha_1(1 - \alpha_1) \alpha_2(1 - \alpha_2) \dots \alpha_{2018}(1 - \alpha_{2018}))^{\frac{1}{2}} \leq \left(\frac{1}{2}\right)^{2018}$$

$$\Rightarrow \text{Product of roots} \leq \frac{1}{2^{2018}}$$

SECTION-I (ii)

7. Ans (D)



$f''(x) = 6x \quad \therefore x_0 = 0$ is point of inflection

8. Ans (D)

$$\begin{aligned} & \int x^3 (\ln x)^2 dx \\ & \Rightarrow (\ln x)^2 \cdot \frac{x^4}{4} - \int 2 \frac{\ln x}{x} \cdot \frac{x^4}{4} dx \\ & \Rightarrow \frac{x^4}{4} (\ln x)^2 - \frac{1}{2} \left[\ln x \cdot \frac{x^4}{4} - \int \frac{1}{x} \cdot \frac{x^4}{4} dx \right] \\ & \Rightarrow \frac{x^4}{4} (\ln x)^2 - \frac{1}{2} \left[\frac{x^4}{4} \ln x - \frac{x^4}{16} \right] + e \\ & \Rightarrow \frac{x^4}{4} (\ln x)^2 - \frac{1}{8} x^4 \ln x + \frac{x^4}{12} + e \\ & \Rightarrow \frac{x^4}{32} [8(\ln x)^2 - 4 \ln x + 1] + e \end{aligned}$$

$a = 8, b = 2, c = -4, d = 1$

9. Ans (B)

We have

$$\int \left(\frac{f''(x) - f'(x)}{f'(x) - f(x)} \right) dx = \int dx$$

$$\Rightarrow f'(x) - f(x) = ae^x$$

$\therefore$ on solving, we get

$$f(x) = xe^x.$$

Now, verify it.

10. Ans (D)

(P) Put $x = \sin \theta$

$$\therefore \int_0^{\pi/2} \frac{d\theta}{1 + \cos \theta} = 1$$

$$(Q) \lim_{x \rightarrow 0} \frac{\int_0^x xe^{t^2} dt}{\left(e^{\frac{x^2}{2}} - 1 \right) \cdot \frac{x^2}{2}}$$

$$2 \lim_{x \rightarrow 0} \frac{\int_0^x xe^{t^2} dt}{x} = 2 \lim_{x \rightarrow 0} \frac{e^{x^2}}{1} = 2$$

$$(R) T_r = \frac{1}{\sqrt{n} \sqrt{r}} = \frac{1}{n(\sqrt{r/n})}$$

$$\therefore \text{Sum} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{4n} \frac{1}{\sqrt{r/n}}$$

$$(S) f'(x) = [(2x)^3 - 2(2x) + 4] \times 2$$

$$-[(x-1)^3 - 2(x-1) + 4]$$

$$f'(1) = 12$$

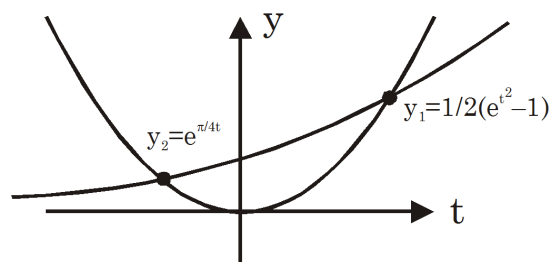
SECTION-II

1. Ans (0.33)

$$e^t \int_0^{\infty} \frac{dx}{(x^2+1)(1+x^{2020})} = e^{t^2} \int_0^t xe^{-x^2} dx$$

$$e^{\frac{\pi}{4}t} = \frac{1}{2} (e^{t^2} - 1)$$

By drawing graph, we get 2 solutions



2. Ans (0.50)

On integrating twice by parts taking e^{-x} as second function we have

$$\begin{aligned} I_m &= [-\sin^m x e^{-x}]_0^\infty - \int_0^\infty m \sin^{m-1} x \cos x (-e^{-x}) dx \\ &= 0 + m \int_0^\infty (\sin^{m-1} x \cos x) e^{-x} dx \\ &= m \left[\left\{ \sin^{m-1} x \cos x \right\} (-e^{-x}) \right]_0^\infty \\ &\quad + m \int_0^\infty [(m-1) \sin^{m-2} x \cos^2 x - \sin^m x] e^{-x} dx \\ &= 0 + m(m-1) \int_0^\infty \sin^{m-2} x (1 - \sin^2 x) e^{-x} dx - m \int_0^\infty e^{-x} \sin^m x dx \\ &= m(m-1) \int_0^\infty e^{-x} \sin^{m-2} x dx - m(m-1) \int_0^\infty e^{-x} \sin^m x dx - m \int_0^\infty e^{-x} \sin^m x dx \\ I_m &= m(m-1) I_{m-2} - m^2 I_m \end{aligned}$$

$$\text{or } (1 + m^2) I_m = m(m-1) I_{m-2} \quad \dots(i)$$

To evaluate put successively in (i), so that

$$17I_4 = 12I_2 \text{ or } I_4 = \frac{12}{17} I_2$$

$$\text{and } 5I_2 = 2I_0, \text{ i.e., } I_2 = \frac{2}{5} I_0$$

$$\text{But } I_0 = [-e^{-x}]_0^\infty = 1$$

$$\therefore I_4 = \frac{12}{17} \times \frac{2}{5} \times 1 = \frac{24}{85}$$

3. Ans (3.00)

We have to minimize area enclosed between y

$$= 4x - x^2, y = k, x = 0, x = 4.$$

Since $y = 4x - x^2$ is symmetric about $x = 2$

so in $[0, 2]$, the minima occurs at $x = 1$

$$\therefore k = 3$$

4. Ans (4.00)

$$x = 3 \cos \theta - \cos^3 \theta \Rightarrow \frac{dx}{d\theta} = -3 \sin^3 \theta$$

$$y = 3 \sin \theta - \sin^3 \theta \Rightarrow \frac{dy}{d\theta} = 3 \cos^3 \theta$$

$$\therefore \frac{dy}{dx} = -\frac{\cos^3 \theta}{\sin^3 \theta}$$

equation of normal is

$$y - (3 \sin \alpha - \sin^3 \alpha) = \frac{\sin^3 \alpha}{\cos^3 \alpha} (x - (3 \cos \alpha - \cos^3 \alpha))$$

$$\Rightarrow y \cos^3 \alpha - x \sin^3 \alpha = \frac{3}{4} \sin^4 \alpha$$

$$\Rightarrow 4 \left[\frac{\cos^4 \alpha}{\cos \alpha} \cdot y - \frac{\sin^4 \alpha}{\sin \alpha} \cdot x \right] = 3 \sin 4\alpha$$

$$\therefore a = \cos^4 \alpha, b = \sin^4 \alpha, a + b = \sin 4\alpha$$

$$\Rightarrow \cos^4 \alpha + \sin^4 \alpha = \sin 4\alpha$$

$$\Rightarrow 2 - \sin^2 2\alpha = 2 \sin 4\alpha$$

$$\Rightarrow 4 - [1 - \cos 4\alpha] = 4 \sin 4\alpha$$

$$\Rightarrow 3 = 4 \cos 4\alpha + 4 \sin 4\alpha$$

$$\Rightarrow \text{Total 4 values of } \alpha$$

5. Ans (1.50)

$$\text{I.F} = e^{-\int \frac{x^4 + 3x^2}{(x^2 + 1)^2} dx} = e^{\frac{-x^3}{x^2 + 1}}$$

$$\Rightarrow y e^{\frac{-x^3}{x^2 + 1}} = \int (4x + 3) dx$$

$$\Rightarrow f(x) = y = (2x^2 + 3x + 1) e^{\frac{x^3}{x^2 + 1}}$$

$$\Rightarrow f(1) = 6\sqrt{e} \text{ \& } f(-1) = 0$$

$$\Rightarrow f(-1) = 0$$

6. **Ans (4.90)**

Let $x \in \left[\frac{1}{n+1}, \frac{1}{n} \right]$ where n is some integer

between 1 and 119 for which $nx - 1 \leq 0$ and

$$(n+1)x - 1 \geq 0$$

Hence,

$$f(x) = (1-x) + (1-2x) + \dots$$

$$+ (1-nx) + ((n+1)x-1) + ((n+2)x-1) + \dots$$

$$+ (118x-1) + (119x-1)$$

$$f(x) = \sum_{k=1}^n (1-kx) + \sum_{k=n+1}^{119} (kx-1)$$

$$= 2 \sum_{k=1}^n (1-kx) + \sum_{k=1}^{119} (kx-1)$$

$$= \{60.119 - n(n+1)\}x + 2n - 119$$

$f(x)$ is increasing on the interval $\left[\frac{1}{n+1}, \frac{1}{n} \right]$ iff.

$$60.119 - n(n+1) > 0. \text{ Therefore,}$$

minimum value of $f(x)$ occurs for $60.119 - n(n+1)$

$$= 0 \text{ which given } n = 84$$

$$\Rightarrow f(84) = (60.119 - 84.85)x + 2(84) - 119 = 49$$

7. **Ans (0.80)**

$$\frac{n^3}{(n+1)^3} = \frac{f(n+1)}{f(n)}$$

$$f(n+1) = \frac{f(1)}{(n+1)^3}$$

$$f(2018) = 1, f(1009) = (2)^3$$

$$\frac{f(1009) - 1}{2} + \int_1^8 ([x] - \{x\}) dx = 28$$

8. **Ans (7.00)**

$$I_7 = \int_0^1 (1-x^4) \cdot \frac{1}{x} dx$$

$$I_7 = \left[x(1-x^4)^7 \right]_0^1 - \int_0^1 7(1-x^4)^6 (-4x^3) \cdot x dx$$

$$I_7 = 0 + 28 \int_0^1 (1-x^4)^6 x^4 dx$$

$$I_7 = 28 \int_0^1 (1-x^4)^6 [1 - (1-x^4)] dx$$

$$I_7 = 28I_6 - 28I_7 \Rightarrow 29I_7 = 28I_6$$

$$\Rightarrow \frac{29I_7}{4I_6} = 7$$